Multi-level Contrastive Learning for Knowledge Tracing

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Knowledge Tracing (KT) is the task of predicting students' future performance based on their past interactions with educational resources. A key aspect of KT is representation learning, which aims to capture meaningful features from students' learning behaviors to improve prediction performance. Recently, contrastive learning methods have shown great promise in representation learning. As a result, KT models based on contrastive learning have been introduced to enhance representation learning for KT. However, these models have posed several challenges. Firstly, most of these models adopt the contrastive learning approach used in other fields, which involves data augmentation followed by contrastive learning, yet effectively applying data augmentation in KT remains an open challenge. Secondly, these models typically apply contrastive learning to only one of the fundamental components of KT: questions, interactions, or knowledge states, thereby limiting their overall performance. To address these issues, this article proposes a Multi-level Contrastive learning model for Knowledge Tracing (MCKT). MCKT (The code can be found at https://github.com/lilstrawberry/MCKT.) does not rely on data augmentation strategies; instead, it deeply integrates domain knowledge and performs contrastive learning at three levels: questions, interactions, and knowledge states. Experimental results on four publicly available datasets, compared against a total of 20 state-of-the-art KT models, demonstrate that MCKT consistently outperforms other models. Subsequent experiments further validate the effectiveness of the multi-level contrastive learning approach.

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1 Introduction

In recent years, with the rapid development of multimedia technology, educational methods have become increasingly diverse. Particularly since the outbreak of the COVID-19 pandemic, online education using the internet as a medium has gained widespread popularity. Online education platforms such as MOOCs and Coursera have attracted a large number of learners. As online education data accumulates, mining the underlying learning patterns is of paramount importance [77]. Once these learning patterns are captured, educators can tailor instruction to address learners' weaknesses, continually improving their abilities, thus achieving a more personalized form of education [56]. **Knowledge Tracing (KT)** plays a crucial role in online education systems, predicting students' future performance based on their past interactions with educational resources. Effectively addressing KT tasks undoubtedly assists educators in better understanding students' learning needs, thereby providing more personalized and targeted educational support [3].

A simple illustration of KT is shown in Figure 1. In this illustration, each question contains one or more relevant concepts, and students practice with different questions. Their knowledge state, which represents their mastery of knowledge, constantly evolves as they progress. KT involves predicting students' future performance when answering questions. A key element in achieving accurate predictions is effective representation learning, which captures meaningful features from students' learning behaviors and provides a foundation for modeling the dynamic progression of their knowledge over time. Clearly, how to trace and represent students' knowledge states is crucial for KT [1, 43, 47]. In addition, in KT, the number of questions is often very high, leading to sparse interaction data between students and questions [6] (interaction here refers to students' responses to questions in KT, typically indicating whether a question was answered correctly or incorrectly). In such cases, simple representations of questions and interactions can result in catastrophic failures of KT models [6, 66]. Therefore, how to represent questions and interactions is of paramount importance [33, 57, 83]. Several KT models [2, 12, 22, 45] have been proposed to address KT problems. For example, the earliest model proposed to address KT was Bayesian KT [11], which utilizes hidden Markov models to trace students' knowledge states for each concept. Subsequently, with the rise of deep learning, deep learning-based KT models like DKT [43] were introduced, where DKT employs recurrent neural networks to trace students' knowledge states. AKT [16] introduces the Rasch [44] method to represent questions and interactions, along with a context-aware attention mechanism to simulate students' forgetting behavior for more effective representation of students' knowledge states. GIKT [74] employs **Graph Convolutional Networks (GCNs)** [26] to capture the relationships between questions and concepts. It also incorporates state evolution and review modules to represent students' knowledge states.

Recently, contrastive learning methods have demonstrated powerful representation learning capabilities by capturing useful information in data through the learning of similarities and differences between different samples or features [8, 20]. Subsequently, research has attempted to introduce

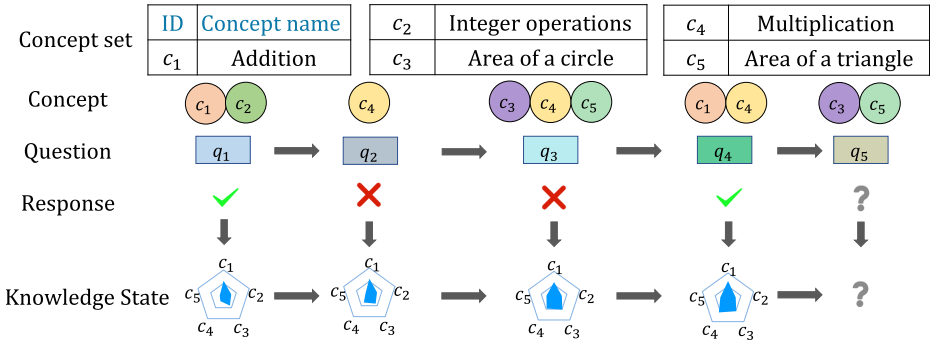


Fig. 1. A simple example of KT. Each question is associated with one or more concepts. A student practices on different questions, continuously accumulating knowledge, and KT involves predicting the probability of the student correctly solving the next question.

contrastive learning methods into the field of KT. For instance, CL4KT [30] proposed various interaction-enhancement strategies, utilizing contrastive learning to construct effective knowledge state representations. Bi-CLKT [51] employed graph-enhancement strategies [78] and employed contrastive learning to learn effective question representations. Currently, contrastive learning-based KT methods are still in the early stages of research. Despite their promising results, they have also brought about some challenges for further research: (1) They often follow contrastive learning paradigms from other domains, i.e., they first perform data augmentation and then apply contrastive learning. However, how to enhance KT through data augmentation strategies has remained an unexplored challenge. For example, data augmentation strategies proposed in CL4KT involve swapping the order of student responses. While this approach, when combined with contrastive learning, can enhance the performance of KT models, it remains unclear whether such practices align with real-world scenarios. Bi-CLKT employs graph-enhancement strategies in the graph domain, such as edge perturbation, which inevitably disrupts the semantic structure of the graph, leading to redundant and unreliable internal connections. (2) They often focus on contrastive learning for only one of the three fundamental elements in KT: questions, interactions, and knowledge states. This limitation results in restricted performance. As mentioned earlier, in KT, the representation of the knowledge state is crucial, and similarly, how to represent the questions and interactions is also very important. Modeling only one of these aspects is clearly insufficient.

For the first challenge, we aim to delve deeper into domain-specific knowledge, achieving the benefits of contrastive learning while breaking free from the limitations of data augmentation. For the second challenge, we believe that using contrastive learning to represent questions, interactions, and knowledge states is beneficial. However, this also brings forth an inquiry: How can we utilize domain-specific knowledge to construct contrastive learning for questions, interactions, and knowledge states? Regarding questions, in fact, in KT, two questions that contain the same concepts can be considered similar; this idea is reflected in much KT research [14, 33, 59]. Therefore, we treat two questions containing the same concepts as positive samples, and otherwise as negative samples. As for interactions, there is currently limited research exploring how to better represent interactions. This is due to the fact that the reasons for answering questions correctly or incorrectly vary from person to person, making it an unresolved issue how to constrain interactions for effective representation. Inspired by a simple intuition that "the reasons for correct answers are often similar, while the reasons for incorrect answers are diverse," we apply contrastive learning to positive interactions (i.e., questions answered correctly). This intuition aligns with Smith et al. [50], who found that correct answers tend to arise from a coherent and shared understanding of concepts, whereas incorrect

answers often reflect diverse and individualized interpretations of prior knowledge. We consider two positive interactions containing the same concepts as positive samples and do not impose constraints on negative interactions (the intuition reflected here is that the representations of two similar questions answered correctly are also similar). Regarding the knowledge state, we have designed two distinct yet complementary knowledge states to contrastive learning based on students' cognition: domain knowledge state and topic knowledge state. The domain knowledge state signifies a student's overall comprehension of knowledge within a particular domain (data in KT is typically targeted towards a specific domain), developed through extensive training. In contrast, the topic knowledge state represents the knowledge state reconstructed by a student specifically for a particular question.

Specifically, in this article, we propose a **Multi-level Contrastive Learning Model for Knowledge Tracing (MCKT)**. It does not rely on data augmentation; instead, it deeply integrates domain knowledge and constructs multi-level contrastive learning from the question-level, interaction-level to the knowledge state-level to learn advantageous representations for KT. MCKT brings together questions containing the same concepts and excludes those without the same concepts, thereby obtaining meaningful and effective question representations for KT. In interaction-level contrastive learning, MCKT brings together positive interaction pairs containing the same concepts, without making any changes to negative interaction pairs. This is because there are various reasons for answering questions incorrectly and it does not restrict it to be more in line with practical significance. In knowledge state-level contrastive learning, we design two different knowledge states for students. The first one is domain knowledge state built based on the entire interaction history, and the second one is topic knowledge state constructed based on reviewing similar past interactions related to the current question. For any student, these two knowledge states characterize their knowledge level to varying degrees. We bring together these two knowledge states of the same student and exclude those of different students. Extensive experiments on four public datasets, with 20 state-of-the-art KT models, show that MCKT significantly outperforms other KT models. We believe that MCKT will provide new insights for KT research, especially for contrastive learning-based KT research.

In summary, our contributions can be summarized as follows:

- —Considering that existing contrastive learning-based KT models rely on data augmentation strategies, we propose a deeper integration of domain knowledge to overcome this limitation.
- —We construct the student's knowledge state from different perspectives, with a particular emphasis on the topic knowledge state that students form by reviewing similar past interactions for the current question. Combining this topic knowledge state with the domain knowledge state enables us to capture the student's knowledge more comprehensively.
- —We propose MCKT, which establishes multi-level contrastive learning from the question-level, interaction-level, to the knowledge state-level, to learn advantageous representations for promoting KT.
- —Extensive experiments on four public datasets involving a total of 20 state-of-the-art KT models consistently demonstrate that MCKT outperforms other models significantly.

The rest of the article is organized as follows: Section 2 provides a review of related work on KT. Section 3 presents the problem statement for KT. In Section 4, we provide a detailed introduction to our MCKT model. Section 5 contains extensive experiments and analyses conducted on MCKT. Section 6 shows an application of KT. Finally, in Section 7, we summarize the article.

2 Related Work

In this section, we first review the fundamental work of KT. Then, we explore two topics relevant to this article: Representation learning for KT and contrastive learning for KT.

2.1 KT

KT can be traced back to the proposal of Bayesian KT [11]. In subsequent work, there have also been some KT methods based on factor analysis introduced [5, 42, 65, 69]. In this article, we focus exclusively on deep learning-based KT models. In recent years, with the rise of deep learning techniques, research in KT based on deep learning has made significant advancements. We categorize KT models into the following four types based on their characteristics: *Sequential based*, *Memory based*, *Graph based*, and *Attention based*.

Sequential Based. In sequential-based KT modeling methods, DKT [43] was the first deep learning-based KT model to use LSTM [23] to trace students' knowledge states and achieved impressive performance. Considering that DKT cannot reconstruct observed inputs and has unstable knowledge states, DKT+ [76] proposed to enhance DKT by introducing reconstruction error and stable knowledge states. As knowledge forgetting is inevitable during the student learning process, DKT+forgetting [38] incorporated forgetting mechanisms into DKT to enhance its performance. ATKT [19] found that deep learning-based KT models have a risk of overfitting and introduced adversarial training [17] to mitigate overfitting. LPKT [46] focuses on maintaining the consistency of students' knowledge states during their learning process, enhancing interpretability while maintaining high performance. DIMKT [45] delves into the impact of question difficulty on students' knowledge states to enhance KT. LBKT [73] explores the influence of students' learning behavior and knowledge states in-depth to strengthen KT.

Memory Based. In memory-based KT modeling methods, DKVMN [79] was the first to propose a Dynamic Key-Value Memory Network to model students' knowledge states, considering that each concept is associated with many potential knowledge concepts. SKVMN [1], based on DKVMN, introduced a Hop-LSTM structure to enhance the representation capability of knowledge states. DGMN [2] considers relationships between latent concepts and takes into account students' forgetting behaviors. MAN [22] addresses skill transition phenomena and proposes a memory-enhanced attention network.

Graph Based. In graph-based KT modeling methods, GKT [39] designed multiple strategies to construct knowledge graph structures and propagated knowledge states on this graph. SKT [63] considered various relationships between concepts on the graph, such as prerequisite and similarity relationships. HGKT [62] proposes a hierarchical graph KT to explore potentially complex relationships between exercises.

Attention Based. In attention-based KT modeling methods, SAKT [40] was the first to use a self-attention mechanism to capture the degree of student attention to concepts in building their knowledge states. SAINT [9] used a Transformer [64] to construct students' knowledge states. AKT [16] proposes a context-aware attention architecture to simulate students' forgetting behavior. sparseKT [24] recognizes the risk of overfitting on small datasets in KT and introduces a sparse attention KT model to enhance KT robustness.

2.2 Representation Learning for KT

In deep learning-based KT, representation learning is crucial for capturing meaningful patterns in students' learning behaviors and improving prediction performance. Based on students' learning processes, KT consists of three fundamental components: questions, interactions, and knowledge states [35, 43, 58].

Questions. Questions refer to the exercises or learning units that students encounter during their learning process. They form the most fundamental component of KT. To enhance question representations, researchers have made numerous efforts: DHKT [66] explores the relationships between questions to constrain their representations. EERNN [53] leverages textual information of

questions, using word2vec [37] to obtain question embeddings. AKT [16], inspired by the Rasch [44] model, positions question representations near relevant concepts. MIKT [58] extends AKT by introducing an extended Rasch model, accommodating multi-concept representations for questions. SimQE [55] employs random walks on a heterogeneous graph with multiple entity relations to learn question embeddings.

Interactions. Interactions capture students' responses to questions, playing a key role in inferring their knowledge mastery. Despite their importance, research on improving interaction representations remains relatively limited. The most common approach is to combine question and response information through specific operations, as implemented in models such as simpleKT [35], LBKT [73], and ReKT [47]. Some studies have explored more sophisticated ways to represent interactions: CAKT [83] enforces similarity constraints on interactions with the same question to improve their representations. MRT-KT [12] introduces a multi-relation transformer to model complex interaction representations.

Knowledge States. Knowledge states represent a student's implicit mastery of knowledge concepts, making them the core of predicting future performance. Numerous approaches have been proposed to model and represent knowledge states effectively: DKT [43] employs LSTMs to trace students' evolving knowledge states. LPKT [46] introduces a stable knowledge acquisition process to constrain knowledge state representations. LBKT [73] incorporates multiple learning behaviors to enhance the representation of knowledge states. ReKT [47] adopts a multi-perspective approach for representing student knowledge states. GRKT [13] applies **Graph Neural Networks (GNNs)** to model the interdependencies among concepts for more accurate knowledge state representations.

In summary, questions, interactions, and knowledge states are the foundational components of KT. Effectively representing them is critical to advancing KT models and improving their predictive performance.

2.3 Contrastive Learning for KT

Contrastive learning is a specialized branch of self-supervised learning [32]. In recent years, contrastive learning has demonstrated strong representation learning capabilities in various fields such as computer vision [8, 18, 20], natural language processing [15, 72], GNNs [61, 78, 82], and more. These studies indicate that customized data augmentation strategies tailored to specific tasks are crucial for the success of contrastive learning. Various data augmentation strategies, such as cropping and rotation for images in computer vision, synonym replacement and word insertion for natural language processing, and feature masking and random edge removal for GNNs, have been shown to be effective.

The KT methods based on contrastive learning are still in their early stages, with only a limited amount of research using contrastive learning methods to promote KT. Bi-CLKT [51] was the first to introduce contrastive learning into KT, using it on both question-question graphs and concept-concept graphs, following the paradigm of graph augmentation-contrastive learning, to obtain representations of questions and concepts. TCL4KT [54] represents questions from three different perspectives on the concept-question-student heterogeneous graph to perform contrastive learning and enhance DKT. $S^2 - HHN$ [71] utilizes multiple data features of KT, considers higher-order and heterogeneous relationships among questions on a heterogeneous graph, and uses contrastive learning to obtain good question representations. CL4KT [30] designs enhancement strategies for questions and responses to build different knowledge states for students based on the AKT [16] base model for contrastive learning. DTransformer [77] combines different sequence augmentation strategies to construct various knowledge states for contrastive learning to ensure stable student knowledge. GraphCA [67] introduces a novel graph counterfactual [27] enhancement method to obtain good user representations through contrastive learning.

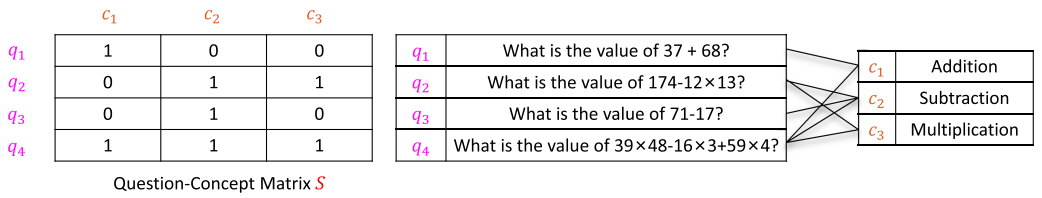


Fig. 2. A simple example of question-concept matrix S. Each question is associated with one or more concepts.

Unlike these contrastive learning-based KT methods, there are several key differences with MCKT. Firstly, MCKT's contrastive learning does not rely on data augmentation strategies, whereas they all require designing various data augmentation strategies for contrastive learning. Although combining contrastive learning has been proven effective in these methods, the impact it brings remains unknown. For example, in methods like CL4KT and DTransformer, they involve random swapping of students' answer sequences and flipping the correctness of student responses. While these approaches align with intuition, their effects on students' experience and learning in real educational settings are still unknown. Secondly, they all focus solely on modeling one of the three fundamental elements of KT: the question, interaction, or knowledge state, disregarding the crucial role of other elements. While methods like TCL4KT [54] represent questions from various perspectives, it still primarily constructs contrastive learning at the question-level. In contrast, MCKT constructs multi-level contrastive learning, encompassing the question-level, interaction-level, and knowledge state-level.

It is worth noting that MCKT can further enhance its performance by integrating graph structures. **Graph Contrastive Learning (GCL)** [4, 68] can capture complex relationships among questions, interactions, and knowledge states by modeling them as graph nodes. Through graph attention mechanisms or other techniques, GCL can focus on relevant connections, improving MCKT's ability to learn contextual and semantic relationships. This integration will boost MCKT's representation learning and performance in modeling student knowledge. We leave this aspect for future work.

3 Problem Statement

The problem of KT can be formalized as follows: Given a sequence of interactions in student u's learning history, denoted as $L_u = \{(q_1, r_1), (q_2, r_2), (q_3, r_3), \ldots (q_t, r_t)\}$, where $u \in \mathbb{U}$, $\mathbb{U}$ represents all students, q_t represents the question that student u needs to answer at time t, and r_t indicates whether student u answered question q_t correctly ($r_t = 1$ for a correct answer, $r_t = 0$ for an incorrect answer). The task of KT is to predict the probability $P(r_{t+1} = 1|L_u, q_{t+1})$ of student u correctly answering the question q_{t+1} at time $t + 1$.

Additionally, it's worth mentioning that each question is associated with one or more concepts. For instance, the question "What is the value of $1 + 22 \times 13$?" encompasses the concepts of "addition" and "multiplication." Formally, we represent the question-concept matrix as $S \in \mathbb{R}^{n \times m}$, where n represents the total number of questions, m represents the total number of concepts. If a specific question $q \in \mathbb{Q}$ is related to a particular concept $c \in \mathbb{C}$, then $S_{qc} = 1$; otherwise, $S_{qc} = 0$. Here, $\mathbb{Q}$ represents the set of all questions, $\mathbb{C}$ represents the set of all concepts, and S_{qc} denotes the element in the qth row and cth column of the matrix S. As illustrated in Figure 2 in this example, where $n = 4$ and $m = 3$, for the question q_2, it is associated with concepts c_2 and c_3.

4 MCKT

In this section, we will provide a detailed overview of the various components of MCKT. The overall architecture of MCKT is depicted in Figure 3. We will start by discussing the model's embedding

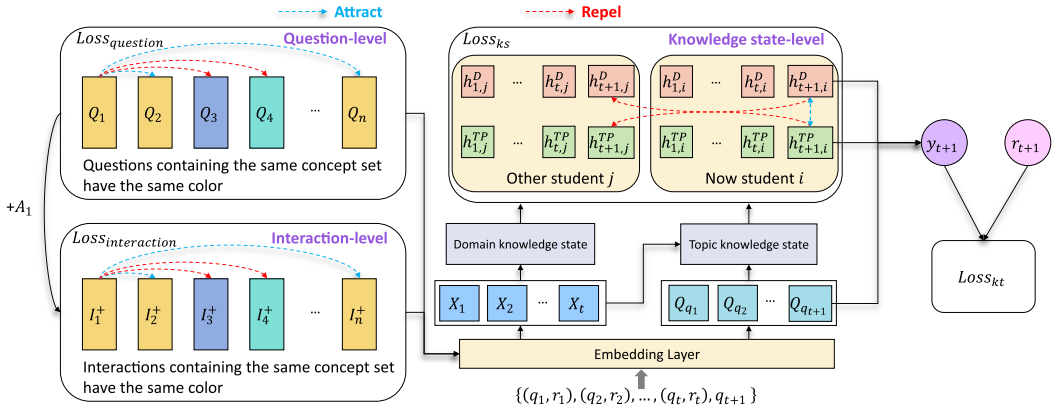


Fig. 3. The overall architecture of MCKT. During the question-level contrastive loss calculation process, we pull in questions that contain the same concept set, otherwise we exclude them. In the calculation process of interaction-level contrast loss, we also pull in the positive interactions containing the same concept set. In the knowledge state-level contrast loss, we pull in the domain knowledge state and topic knowledge state of the same student, and exclude the domain knowledge state and topic knowledge state of different students. Finally, we combine the current question with the student's domain and topic knowledge state to obtain the final prediction of the model.

component, followed by a detailed introduction of MCKT's question-level, interaction-level, and knowledge state-level contrastive learning. Finally, we will explain how MCKT predicts a student's response based on their knowledge state. Furthermore, to better illustrate MCKT, we will provide a summary of the entire algorithmic process of MCKT.

4.1 Embedding

MCKT's input features consist of questions, concepts, and responses. We denote the embedding matrix for questions as $Q \in \mathbb{R}^{n \times d}$, the embedding matrix for concepts as $C \in \mathbb{R}^{m \times d}$, and the embedding matrix for responses as $A \in \mathbb{R}^{2 \times d}$, where n represents the total number of questions, m represents the total number of concepts, and d represents the dimensionality of the embedding features. Q_i, C_i, and A_i, respectively, refer to the ith row of Q, C, and A.

Assuming the current timestep is t, the interaction (q_t, a_t) of the student at the current timestep t after embedding is represented as follows:

$$X_t = Q_{q_t} + A_{a_t}, \tag{1}$$

where X_t represents the interaction embedding at time t, Q_{q_t} denotes the q_tth row of Q, and A_{a_t} represents the a_tth row of A. Please note that some KT methods [40, 43, 79] are known to use an embedding matrix of size $\mathbb{R}^{2n \times d}$ to represent interactions, where this matrix is designed to differentiate between correct and incorrect responses to a particular question. However, a drawback of this approach is that it overlooks the fact that both cases involve the same question. In contrast, while the interaction embeddings in MCKT are quite simple, they effectively capture this characteristic. Furthermore, if we learn an effective representation of the questions, this interaction embedding will also benefit from it.

For the next timestep $t + 1$, when the student needs to answer the question q_{t+1}, we represent it in embedded form as $Q_{q_{t+1}}$.

Clearly, the embedding component of MCKT is very straightforward. This is because we aim to simplify this aspect and focus on the core of MCKT (multi-level contrastive learning). This also helps subsequent KT research to more easily grasp the core part of MCKT.

4.2 Question-Level Contrastive Learning

The main purpose of question-level contrastive learning is to acquire better question representations to enhance KT. Numerous studies [16, 33, 66, 74] have shown that exploring the relationships between questions and concepts is crucial for improving question representations. Some representative methods, such as the Rasch representation proposed by AKT [16], explicitly constrain the representation of questions to their associated concepts, while GIKT [74] proposed using GCNs on question-concept graphs to aggregate their relationships. Although these methods have made improvements in question representation, they are limited to the connections between questions and concepts and fail to fully exploit the inherent relationships between questions. In contrast, contrastive learning emphasizes the similarity and dissimilarity between samples, providing potential opportunities for improving question representations.

In this article, we employ contrastive learning to obtain improved question representations by considering pairs of questions. Specifically, for any two distinct questions $i, j \in \mathbb{Q}$, if $S_i = S_j$, then questions i and j are considered positive samples; otherwise, they are considered negative samples, where S_i represents the ith row of the question-concept matrix S, and S_j represents the jth row of S. We denote the positive samples for question i as pos_i, then:

$$pos_i = \cup j, \ where \ j \in \mathbb{Q} \ and \ S_i = S_j, \tag{2}$$

where $\cup$ represents the union operation.

Then, for any question $i \in \mathbb{Q}$, we pull it closer to its positive samples and push it away from its negative samples, and the contrastive loss for question i is calculated as:

$$Loss^i_{question} = -log \frac{\sum_{k \in pos_i} \exp(sim(Q_i, Q_k)/\tau)}{\sum_{j \in \mathbb{Q}} \exp(sim(Q_i, Q_j)/\tau)}, \tag{3}$$

among them, exp represents the exponential function, sim represents the similarity function, the cosine similarity is used in this article, and τ represents the temperature coefficient, which is a hyperparameter that affects the way the distance between samples is measured.

Finally, the question-level contrastive loss $Loss_{question}$ is calculated as:

$$Loss_{question} = \sum_{i \in \mathbb{Q}} Loss^i_{question}. \tag{4}$$

We will apply this loss to learn good question representations.

4.3 Interaction-Level Contrastive Learning

The goal of interactive-level contrastive learning is to learn good interaction representations to facilitate KT. It's worth mentioning that interaction representations are crucial for constructing a student's knowledge state, and a good knowledge state is essential for the success of KT. Therefore, acquiring good interaction representations is of utmost importance. Such as DKT [43], the pioneering work of KT based on deep learning, which only requires interaction representations as input for constructing knowledge states. However, despite the importance of interaction representations, there is limited research in KT methods on how to represent them, as the current mainstream approach in KT methods involves designing complex model architectures for knowledge state construction.

We aim to learn good interaction representations to enhance KT. Inspired by a simple intuition that "the reasons for correct answers are often similar, while the reasons for incorrect answers are diverse," in this article, we impose constraints on positive interactions and do not impose constraints on negative interactions. For any question $i \in \mathbb{Q}$, its positive interaction (i.e., questions answered correctly) I_i^+ is:

$$I_i^+ = Q_i + A_1, \tag{5}$$

where A_1 represents the embedding representation for correctly answered questions (A_0 represents the embedding representation for incorrectly answered questions). It is important to note that the difference here compared to Equation (1) is that Equation (1) refers to interaction representations at time t, while this equation refers to positive interaction representations (without a time factor and specifically positive interactions).

Notice that since the reasons for students answering questions correctly are quite similar, the reasons for them answering similar questions correctly should also be similar. In other words, for any two distinct positive interactions, if the set of concepts covered by the questions represented in the interaction is the same, then these two positive interactions are considered similar. Clearly, for any question $i \in \mathbb{Q}$, its positive interaction has the same positive sample set as question i, which is pos_i. Similarly, for the positive interactions of any question $i \in \mathbb{Q}$, we bring them closer to their positive samples while pushing them away from their negative samples, the contrastive loss of the positive interaction behind question i is calculated as:

$$Loss^i_{interaction} = -log\frac{\sum_{k \in pos_i} \exp(sim(I_i^+, I_k^+)/\tau)}{\sum_{j \in \mathbb{Q}} \exp(sim(I_i^+, I_j^+)/\tau)}. \tag{6}$$

Finally, the interaction-level contrastive loss $Loss_{interaction}$ is calculated as:

$$Loss_{interaction} = \sum_{i \in \mathbb{Q}} Loss^i_{interaction}. \tag{7}$$

We will apply this loss to learn good interaction representations.

4.4 Knowledge State-Level Contrastive Learning

In KT, the key to predicting students' responses lies in their knowledge state, and it can be said that the representation of the knowledge state is crucial for the success of KT. In this article, we will trace students' knowledge state from two different perspectives: one is the domain knowledge state constructed from the student's entire interaction history, and the other is the topic knowledge state constructed from the student's responses to specific questions. Furthermore, to better harness the potential of these two knowledge states, we employ contrastive learning to understand their similarities and differences.

4.4.1 Domain Knowledge State. Domain knowledge state is designed to trace and represent students' overall mastery of knowledge in the specific domain after long-term learning. Formally speaking, for the student's interaction X_t at time t and the domain knowledge state h_t^D at time t, the student's domain knowledge state h_{t+1}^D at the next moment (that's the time to answer the question) is calculated as:

$$f_{t+1} = \sigma(W_f[h_t^D \oplus X_t] + b_f), \tag{8}$$

$$i_{t+1} = \sigma(W_i[h_t^D \oplus X_t] + b_i), \tag{9}$$

$$\widetilde{C_{t+1}} = tanh(W_C[h_t^D \oplus X_t] + b_C), \tag{10}$$

$$C_{t+1} = f_{t+1} * C_t + i_{t+1} * \widetilde{C_{t+1}}, \tag{11}$$

$$o_{t+1} = \sigma(W_o[h_t^D \oplus X_t] + b_o), \tag{12}$$

$$h_{t+1}^D = o_{t+1} * tanh(C_{t+1}). \tag{13}$$

In the above equations, σ represents the Sigmoid function, tanh represents the Tanh function, and W_f, W_i, W_C, W_o, b_f, b_i, b_C and b_o are all learnable parameters.

In other words, domain knowledge state is updated step by step. In fact, this is very consistent with the general situation of students' learning: step by step accumulation.

4.4.2 Topic Knowledge State. The topic knowledge state is mainly designed for students to solve specific problems. It aims to more accurately extract information that is beneficial to solving problems from the students' interaction history. Generally, when a student needs to answer a specific question, he/she will recall similar questions he/she has done before to reconstruct his/her knowledge state based on his/her cognition. However, this thinking mode often depends on the individual's cognitive level. According to Cognitive Load Theory [60], students with a strong knowledge base experience lower cognitive load, enabling them to process information more efficiently. This means that students with better knowledge can recall similar questions more accurately, which in turn helps them solve problems more effectively. In contrast, students with poorer knowledge face higher cognitive load, which makes it difficult for them to learn from interactive history tasks, as they struggle to process and integrate new information. Therefore, it is difficult to find key information in the interaction history based on the student's knowledge state. Considering that the degree of similarity between questions exists objectively, it will not be affected by personal cognitive level. Therefore, in this article, we find effective information in the interaction history based on the similarity between questions.

Specifically, for the question q_{t+1} that students need to answer at the next moment, its embedding is expressed as $Q_{q_{t+1}}$, and its similarity to the question Q_{q_i} at the previous moment $i \in [1, t+1]$ is $sim(Q_{q_{t+1}}, Q_{q_i})$, where sim represents the cosine similarity function.

However, it is crucial to recognize that if a student were to focus on all their interaction history to solve a specific question, it would hinder rather than aid problem-solving, as it would lead to an overwhelming and diverse array of information to consider. This approach also does not align with real-world scenarios, where students typically extract only key information rather than all available information. Recent research, such as sparseKT [24], suggests that sparse attention is more beneficial for enhancing the robustness of KT. Therefore, we introduce a similarity threshold δ between questions, allowing students to concentrate only on questions with similarity scores surpassing δ.

We denote the historical moments that the student needs to pay attention to as $qmask_{t+1}$. Then:

$$qmask_{t+1} = \cup i, where\ sim(Q_{q_{t+1}}, Q_{q_i}) >= \delta\ and\ i \in [1, t+1]. \tag{14}$$

We use the Transformer [64] architecture to simulate the process by which students build the topic knowledge state from interactions related to these historical moments, as shown in Figure 4.

Typically, given the $query_{t+1}$ at time $t+1$, along with key_i for any timestep $i \in qmask_{t+1}$ (only allowing students to focus on relevant historical moments), attention can be calculated as follows:

$$Attn_{t+1,i} = Softmax\left(\frac{query_{t+1} \cdot key_i^T}{\sqrt{d_k}}\right), \tag{15}$$

where d_k represents the dimensionality of the $query_{t+1}$ and key_i. However, in KT, simply using this method to compute attention scores performs poorly because students often forget when recalling

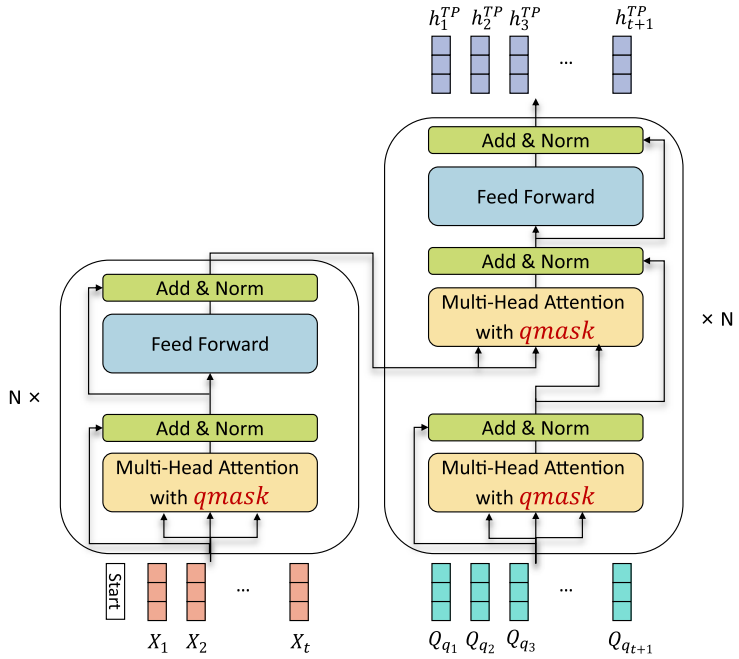


Fig. 4. The overall structure of the topic knowledge state. The way it is constructed differs from the typical Transformer-based KT models in that its attention calculation depends on a *qmask*. The *qmask* ensures that the student only focuses on historical information that is very similar to the current question q_{t+1}, rather than all historical information. The calculation method of *qmask* is shown in Equation (14).

knowledge. As time passes, a student's memory of the knowledge tends to become more vague, and considering the student's forgetting behavior is crucial in KT [2, 38, 46]. Therefore, following the approach of AKT [16], we employ a context-aware attention calculation method:

$$Attn_{t+1,i} = Softmax\left(e^{-\theta \cdot dis(t+1,i)} \frac{query_{t+1} \cdot key_i^T}{\sqrt{d_k}}\right), \tag{16}$$

where θ is a learnable decay rate parameter and $dis(t+1,i)$ is temporal distance measure between timesteps $t+1$ and i. $dis(t+1,i)$ can be calculated as:

$$dis(t+1,i) = |t+1-i| \cdot \sum_{j=i+1, j \in qmask_{t+1}}^{t+1} b_{t+1,j}, \tag{17}$$

$$b_{t+1,j} = \frac{e^{query_{t+1} \cdot key_j^T / \sqrt{d_k}}}{\sum_{g=1}^{t+1} e^{query_{t+1} \cdot key_g^T / \sqrt{d_k}}}. \tag{18}$$

In the Transformer encoder part, the input is $\xi = (X_1, X_2, X_3, \ldots, X_t)$, and in the decoder part, the input is $\eta = (Q_{q_1}, Q_{q_2}, Q_{q_3}, \ldots, Q_{q_{t+1}})$. Other details such as residual connections [21], layer normalization, multi-head attention, and feedforward neural networks are consistent with the Transformer-based KT approach [9, 16, 49, 81]. We record this process as ψ, then the topic knowledge state is calculated as:

$$h_1^{TP}, h_2^{TP}, h_3^{TP}, ..., h_{t+1}^{TP} = \psi(\xi, \eta). \tag{19}$$

The fundamental distinction from the Transformer-based KT approach lies in the construction of h_{t+1}^{TP}. It selects key (relevant) historical moments while considering the forgetting factor, whereas other methods simply focus on all timestep information.

4.4.3 Knowledge State Contrastive Learning. It can be observed that domain knowledge state and topic knowledge state each characterize a student's level of knowledge from different perspectives. However, simply combining the two may not fully harness their effectiveness. For example, when a student answers a question, the domain knowledge state represents the student's overall knowledge level, which is unrelated to the specific question, while the topic knowledge state reflects the student's grasp of that particular question, which is closely related to the specific question. Clearly, combining the two can provide a more comprehensive representation of a student's knowledge state. However, if students can learn the similarities and differences between these two knowledge states (meaning that students can integrate their understanding of domain knowledge and their understanding of specific questions to compare and contrast), this indicates that students can synthesize and apply their knowledge effectively. This contributes to their ability to solve problems.

In this article, we use contrastive learning to capture the similarities and differences between these two knowledge states. It should be noted that although these two knowledge states are different, for the same student, they both reflect the student's understanding of knowledge from different perspectives. That is, for the same student (for example, A), A's domain knowledge state and A's topic knowledge state should be similar, but for different students (for example, A and B), A's domain knowledge state and B's topic knowledge state are different (similarly, A's topic knowledge state and B's domain knowledge state are also distinct).

Specifically, for any student $i \in \mathbb{U}$, we define the domain knowledge state and topic knowledge state of student i at timestep $t+1$ (that's the time to answer the question) as $h_{t+1,i}^{D}$ and $h_{t+1,i}^{TP}$, respectively. The knowledge state contrastive loss for student i at timestep $t+1$, denoted as $Loss_{ks}^{t+1,i}$, is calculated as follows:

$$Loss_{ks}^{t+1,i} = -log\frac{\exp(sim(h_{t+1,i}^{D}, h_{t+1,i}^{TP})/\tau)}{\sum_{j\in\mathbb{U}}\exp(sim(h_{t+1,i}^{D}, h_{t+1,j}^{TP})/\tau)} - log\frac{\exp(sim(h_{t+1,i}^{TP}, h_{t+1,i}^{D})/\tau)}{\sum_{j\in\mathbb{U}}\exp(sim(h_{t+1,i}^{TP}, h_{t+1,j}^{D})/\tau)}, \tag{20}$$

then the knowledge state contrastive loss for all students at time $t+1$ is calculated as:

$$Loss_{ks}^{t+1} = \sum_{i\in\mathbb{U}} Loss_{ks}^{t+1,i}. \tag{21}$$

Finally, the knowledge state contrastive loss $Loss_{ks}$ is calculated as:

$$Loss_{ks} = \sum_{t=0}^{F-1} Loss_{ks}^{t+1}, \tag{22}$$

where F represents the total timesteps for students to answer questions. We will use this loss to accurately represent the student's knowledge state.

4.5 Prediction and Training

For the question $Q_{q_{t+1}}$ that the student needs to answer, as well as his/her two knowledge states h_{t+1}^{D} and h_{t+1}^{TP}, we use the following equation to obtain the model's final prediction result y_{t+1}:

$$y_{t+1} = \sigma(W_2(W_1[h_{t+1}^{D}, h_{t+1}^{TP}, Q_{q_{t+1}}] + b_1) + b_2), \tag{23}$$

among them, W_1, W_2, b_1, b_2 are all learnable parameters. Note that here we directly concat rather than use attention to distinguish the two knowledge states, mainly because these two knowledge states portray students' knowledge from different perspectives, so they play different roles.

Algorithm 1: Training Process of MCKT

Require: Student interaction history L_u, question-concept matrix S, embedding feature dimension size d, temperature coefficient τ, similarity threshold δ, total time steps F, and the balance coefficients φ_1 and φ_2 for the total loss.

Initialize MCKT parameters ϕ.
while not converage **do**
  Calculate the question-level contrastive loss $Loss_{question}$ using Equation 4.
  Calculate the interaction-level contrastive loss $Loss_{interaction}$ using Equation 7.
  for $t = 0$ to $F - 1$ **do**
    Calculate the domain knowledge state h^D_{t+1} using Equation 13.
    Calculate the topic knowledge state h^{TP}_{t+1} using Equation 19.
    Calculate $Loss^{t+1}_{ks}$ using Equation 21.
  end for
  Calculate the knowledge state-level contrastive loss $Loss_{ks}$ using Equation 22.
  Calculate the knowledge tracing loss $Loss_{kt}$ using Equation 24.
  Calculate the total loss $Loss_{total}$ using Equation 25.
  Update the model parameters ϕ using the Adam algorithm.
end while
return ϕ

Finally, the loss of KT $Loss_{kt}$ is defined as the binary cross-entropy loss between the model's predicted probability y_{t+1} and the student's true answer r_{t+1}:

$$Loss_{kt} = -\sum_{t=0}^{F-1} r_{t+1} log(y_{t+1}) + (1 - r_{t+1}) log(1 - y_{t+1}). \tag{24}$$

Then the final total loss $Loss_{total}$ is calculated as:

$$Loss_{total} = Loss_{kt} + \varphi_1 * (Loss_{question} + Loss_{interaction}) + \varphi_2 * Loss_{ks}. \tag{25}$$

Both φ_1 and φ_2 are hyperparameters used to regulate the balance of loss terms. It's worth noting that we have merged the loss terms for $Loss_{question}$ and $Loss_{interaction}$ into a single term. This decision was made due to their similar effects and to reduce the number of hyperparameters. Finally, we optimize the model parameters using the Adam [25] algorithm.

To provide a better explanation of MCKT, the entire process of MCKT can be seen in Algorithm 1.

5 Experiments

In this section, we will conduct extensive experiments to answer the following questions:

— *RQ1*: How does MCKT perform in predicting student performance compared to state-of-the-art KT methods?
— *RQ2*: How do different components of MCKT influence its performance?
— *RQ3*: How sensitive is MCKT to hyperparameters?
— *RQ4*: What is the stability of MCKT in predicting student performance?
— *RQ5*: What is the time complexity of MCKT?

5.1 Datasets

We evaluated the performance of MCKT on the following four publicly available datasets:

Table 1. Summary Statistics of Processed Datasets

	ASSIST09	ASSIST17	Statics2011	EdNet
# Student	4,160	1,706	331	5,000
# Question	15,680	1,150	633	11,775
# Concept	167	86	106	1,837
# Interaction	207,659	153,324	91,133	1,156,254

— *ASSIST09*:[1] Collected from the ASSISTments online educational platform during 2009–2010.
— *ASSIST17*:[2] Collected from the ASSISTments online educational platform in 2017.
— *Statics2011*:[3] Collected from a college-level engineering course on statics.
— *EdNet*:[4] A dataset collected by Santa [10], an online tutoring platform, from 2017 to 2019.

Based on previous research, for the ASSIST series datasets, we removed scaffold questions and records without concepts [16]. In the case of Statics2011, original question indexes and step indexes were combined into a new question index. If the same question was answered consecutively, only the first answer was retained. Moreover, due to the large scale of EdNet datasets, and limitations in computational resources, similar to previous work [29, 33, 74], we randomly sampled records from 5,000 students. The statistical information for these datasets is provided in Table 1.

5.2 Baselines

To assess the performance of MCKT, we compared it with 20 state-of-the-art KT models as follows:

— *BKT* [11]: Traces students' knowledge states using hidden Markov models.
— *DKT* [43]: Traces students' knowledge states using LSTM [23].
— *DKT+* [76]: Enhances DKT by addressing inconsistent knowledge states and irrecoverable inputs.
— *KQN* [28]: Predicts students' performance using knowledge state and concept encoder.
— *DKT+forgetting* [38]: Augments DKT by incorporating various behavioral features to consider forgetting in student knowledge states.
— *PEBG+DKT* [33]: Enhances DKT by deeply exploring the relationship between questions and concepts to obtain pre-trained question representations.
— *ATKT* [19]: Use adversarial training to improve the generalization ability of the model.
— *CDKT* [14]: On the basis of DKT, use contrastive learning between questions to represent questions.
— *DIMKT* [45]: Traces the impact of question difficulty on students' knowledge states.
— *QIKT* [6]: Models centered around questions and enhances model interpretability with interpretable parameters.
— *AT-DKT* [34]: Enhances DKT with two additional tasks to improve performance.
— *DKVMN* [79]: Traces students' knowledge states using a dynamic key-value memory network.
— *Deep-IRT* [75]: Integrates IRT with DKVMN to improve interpretability.
— *GKT* [39]: Propagates students' conceptual knowledge states using a graph structure.
— *SAKT* [40]: Captures student-concept relationships using self-attention mechanism.

[1] https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data.
[2] https://sites.google.com/view/assistmentsdatamining/dataset.
[3] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507.
[4] https://github.com/riiid/ednet.

— *SAINT* [9]: Employs a Transformer [64] architecture to model students' knowledge states.
— *AKT* [16]: Simulates students' forgetting behavior using context-based attention mechanism.
— *CL4KT* [30]: Mitigates sparsity in student-concept interactions using contrastive learning.
— *simpleKT* [35]: Simplifies the model structure based on AKT, achieving simplicity without sacrificing performance.
— *DTransformer* [77]: Uses contrastive learning to trace students' stable knowledge states.

Please note that, in order to ensure a fair comparison of the models, we have not included state-of-the-art KT methods such as SAINT+ [49], SKT [63], LPKT [46], DGMN [2], LBKT [73], EKT [31], and others that utilize additional features such as student response times, question text information, and so on. Consistent with most KT research, we have only included state-of-the-art KT models that take input consisting solely of concepts, questions, and responses (including MCKT).

5.3 Experimental Setup

We implemented MCKT using Pytorch [41] with the following settings: a learning rate of 0.002, a batch size of 80, a temperature coefficient τ of 0.8. Additionally, L2 weight regularization with a weight decay of 1e-5 was applied to the model's weights. To prevent overfitting, we set the dropout rate to 0.4. For data preprocessing, sequences with a length less than 3 were removed from the dataset. Since input sequence lengths varied, we standardized all sequences to a fixed length of 200 (i.e., total timesteps $F = 200$). The embedding dimension size $d \in [16, 32, 64, 128, 256]$, the similarity threshold $\delta \in [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1]$, and the balance coefficients for the total loss $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$. We will discuss their impact on MCKT in Section 5.6. For each dataset, we used 80% of all sequences as the training set and 20% as the test set [48, 74, 77]. We conducted five experiments and reported the average results [2, 53]. All models are trained on a Linux server with two 2.00 GHz Intel(R) Xeon(R) CPUs and a Nvidia Tesla P100-PCIE-16GB GPU. Consistent with prior work [7, 36, 52, 70, 80], we use **Area Under the Curve (AUC)** as the first evaluation metric and ACC (Accuracy) as the second metric. Higher values for both metrics indicate superior model performance.

5.4 Performance (RQ1)

Table 2 presents a performance comparison of MCKT with other KT baselines, with the best performance highlighted in bold and the second-best performance indicated with an underline. The grouping details of the models are discussed in Section 2. BKT is a machine learning-based model (all others are deep learning-based models) and is thus not included in this grouping. The construction of MCKT's domain and topic knowledge states involves sequence and attention modeling, respectively, so its grouping is Sequential based+Attention based. Based on Table 2, we can draw the following conclusions: (1) Deep learning-based KT models significantly outperform machine learning-based KT models (BKT), demonstrating the effectiveness of deep learning in KT. (2) Sequential-based models like DKT and QIKT effectively capture the temporal evolution of student knowledge and have shown strong performance in early KT research. However, they rely on a single evolving knowledge state, which makes them less effective in modeling recent answering behaviors. MCKT introduces a topic knowledge state, allowing it to focus on question-topic-specific answering patterns, ensuring better adaptability to recent interactions. (3) Attention-based models such as simpleKT and DTransformer improve KT by dynamically weighting past interactions, allowing them to focus on important historical events. However, they lack a stable long-term knowledge representation, making them sensitive to noisy interactions. MCKT integrates a domain knowledge state, which aggregates long-term learning history, ensuring a more robust and holistic representation of student proficiency. (4) KT methods incorporating contrastive learning, such

Table 2. Comparison of MIKT and 20 KT Models on Four Datasets

Model	Model Group	ASSIST09		ASSIST17		Statics2011		EdNet	
		AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
BKT [11]	-	0.7180	0.6922	0.6519	0.6290	0.7444	0.7973	0.6737	0.7018
DKT [43]	Sequential based	0.7684	0.7297	0.7188	0.6673	0.8483	0.8275	0.7006	0.7129
DKT+ [76]	Sequential based	0.7783	0.7337	0.7095	0.6622	0.8718	0.8270	0.7028	0.6698
KQN [28]	Sequential based	0.7546	0.7249	0.7065	0.6611	0.8031	0.7969	0.6909	0.7117
DKT+forgetting [38]	Sequential based	0.7717	0.7295	0.7165	0.6665	0.8467	0.8182	0.7018	0.7159
PEBG+DKT [33]	Sequential based	0.7738	0.7329	0.7619	0.6949	0.8655	0.8368	0.7571	0.7366
ATKT [19]	Sequential based	0.7735	0.7332	0.7198	0.6699	0.8175	0.7991	0.7027	0.7109
CDKT [14]	Sequential based	0.7733	0.7297	0.7709	0.7019	0.8872	0.8510	0.7645	0.7386
DIMKT [45]	Sequential based	0.7704	0.7310	0.7682	0.6993	0.8897	0.8501	0.7623	0.7368
QIKT [6]	Sequential based	0.7801	0.7377	0.7645	0.6985	0.8817	0.8482	0.7579	0.7327
AT-DKT [34]	Sequential based	0.7671	0.7293	0.7265	0.6702	0.8687	0.8386	0.7039	0.7136
DKVMN [79]	Memory based	0.7629	0.7266	0.7142	0.6639	0.8363	0.8224	0.6975	0.7120
Deep-IRT [75]	Memory based	0.7657	0.7279	0.7174	0.6647	0.8408	0.8262	0.6997	0.7124
GKT [39]	Graph based	0.7666	0.7290	0.7203	0.6685	0.8384	0.8224	0.6943	0.7104
SAKT [40]	Attention based	0.7564	0.7192	0.7079	0.6596	0.8092	0.8111	0.6956	0.7115
SAINT [9]	Attention based	0.7515	0.7134	0.7537	0.6894	0.8279	0.8110	0.7621	0.7370
AKT [16]	Attention based	0.7850	0.7429	0.7572	0.6916	0.8718	0.8376	0.7616	0.7372
CL4KT [30]	Attention based	0.7626	0.7275	0.7139	0.6615	0.8251	0.8170	0.6965	0.7118
simpleKT [35]	Attention based	0.7772	0.7315	0.7570	0.6899	0.8614	0.8350	0.7627	0.7373
DTransformer [77]	Attention based	0.7646	0.7223	0.7538	0.6898	0.8686	0.8513	0.7501	0.6954
MCKT	Sequential based+Attention based	**0.7950***	**0.7474***	**0.7765***	**0.7071***	**0.8975***	**0.8586***	**0.7728***	**0.7441***

Best results in bold, next best underlined.
*Indicates t-test p-value < 0.05 compared to the second-best result. Model grouping details can be found in Section 2 (Related Work).

as CDKT and CL4KT, enhance representation learning by differentiating similar and dissimilar fundamental components of KT. However, they typically apply contrastive learning at a single level (e.g., question-level or knowledge state-level), which limit their ability to capture different aspects of student learning interactions. MCKT extends contrastive learning to three levels—question, interaction, and knowledge state—allowing for a more structured and comprehensive representation of student learning behaviors. (5) CDKT and AKT demonstrate strong baseline performance. CDKT benefits from the effective question representation based on contrastive learning, while AKT leverages its Rasch-based representation, which models the relationship between questions and concepts. Additionally, its context-aware attention mechanism effectively simulates students' forgetting behaviors. (6) MCKT achieved the highest AUC and ACC across all datasets, with statistically significant improvements. This provides strong evidence for the effectiveness of the multi-level contrastive learning approach proposed in this article.

5.5 Effect of Different Components (RQ2)

In this section, we will analyze how different components of MCKT impact its performance. Specifically, we will start with conducting ablation experiments to investigate the effectiveness of multi-level contrastive learning. Secondly, we will analyze how different knowledge states affect the performance of MCKT.

5.5.1 Ablation Study. We conducted ablation experiments on MCKT to explore the effectiveness of its multi-level contrastive learning. As shown in Table 3, where "Q" indicates the inclusion of question-level contrastive loss, "I" indicates the inclusion of interaction-level contrastive loss, and "KS" indicates the inclusion of knowledge state-level contrastive loss, the best-performing results are highlighted in bold. We can draw the following conclusions: (1) Performance is significantly worse when no contrastive loss is included compared to other scenarios, indicating that the

Table 3. Ablation Experiments for Multi-Level Contrastive Learning in MCKT

Q	I	KS	ASSIST09		ASSIST17		Statics2011		EdNet	
			AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
			0.7671	0.7297	0.7684	0.7004	0.8931	0.8541	0.7716	0.7423
√			0.7848	0.7443	0.7696	0.7013	0.8943	0.8535	0.7721	0.7425
	√		0.7891	0.7456	0.7706	0.7015	0.8941	0.8551	0.7719	0.7426
		√	0.7680	0.7299	0.7752	0.7056	0.8960	0.8551	0.7725	0.7433
√	√		0.7929	0.7465	0.7714	0.7019	0.8945	0.8532	0.7720	0.7430
√		√	0.7869	0.7453	0.7754	0.7064	0.8962	0.8549	0.7718	0.7425
	√	√	0.7907	0.7463	0.7752	0.7048	0.8951	0.8554	0.7714	0.7424
√	√	√	**0.7950**	**0.7474**	**0.7765**	**0.7071**	**0.8975**	**0.8586**	**0.7728**	**0.7441**

"Q" indicates whether question-level contrastive loss is included, "I" indicates whether interaction-level contrastive loss is included, and "KS" indicates whether knowledge state-level contrastive loss is included. The best results are highlighted in bold.

proposed contrastive losses in this article indeed contribute to KT. (2) Including only one of the contrastive loss levels results in a noticeable performance improvement compared to not including any contrastive loss. In particular, question-level and interaction-level contrastive losses lead to the most substantial improvements on datasets like ASSIST09, possibly because this dataset is sensitive to the representation of questions and interactions, and effectively representing them greatly benefits the model. Similarly, on the ASSIST17 dataset, the inclusion of knowledge state-level contrastive loss leads to a more pronounced improvement, indicating that the mutual contrastive learning of domain and topic knowledge states is indeed beneficial. (3) When any two levels of contrastive loss are included, performance improves compared to scenarios with only one level or none. This suggests that the various levels of contrastive loss are reasonable, and combining them genuinely promotes the model's performance from different perspectives. (4) MCKT, which includes all levels of contrastive loss, achieves the best performance, demonstrating the effectiveness of the multi-level contrastive losses proposed by MCKT.

5.5.2 The Influence of Knowledge State. In this section, we will explore the impact of different knowledge states on MCKT. Specifically, we explore the effect of including only one of MCKT's knowledge states. As shown in Table 4, where MCKT-D represents MCKT using only domain knowledge state, MCKT-TP represents MCKT using only topic knowledge state (without knowledge state-level contrastive loss). Additionally, we added MCKT-w/o forget to explore the impact of forgetting on knowledge states. MCKT-w/o forget removes the forgetting of the knowledge state (i.e., Equation (16) is modified to Equation (15)). Based on Table 4, we can draw the following conclusions: (1) A direct comparison between MCKT-D and MCKT-TP reveals that MCKT-D exhibits a significant performance advantage. This is expected since constructing these two knowledge states involves different amounts of data. MCKT-D utilizes the entire interaction history, while MCKT-TP only uses interactions with the current question that surpass the similarity threshold δ. This observation aligns with real-world scenarios where students who solely focus on solving specific questions tend to perform poorly because they often concentrate on these specific questions while neglecting other critical aspects that contribute to problem-solving, such as accumulated problem-solving strategies and skills. In other words, they fail to integrate their knowledge effectively. MCKT outperforms both of these scenarios, indicating the effectiveness of the two proposed knowledge states, and combining them contributes to a more comprehensive representation of

Table 4. The Impact of Different Knowledge States on MCKT Performance

Model	ASSIST09		ASSIST17		Statics2011		EdNet	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
MCKT-D	0.7924	0.7456	0.7718	0.7018	0.8913	0.8529	0.7660	0.7390
MCKT-TP	0.7603	0.7201	0.7115	0.6541	0.8541	0.8288	0.7385	0.7269
MCKT-w/o forget	0.7932	0.7457	0.7734	0.7022	0.8929	0.8545	0.7709	0.7412
MCKT	**0.7950**	**0.7474**	**0.7765**	**0.7071**	**0.8975**	**0.8586**	**0.7728**	**0.7441**

"MCKT-D" represents using only domain knowledge states, "MCKT-TP" represents using only topic knowledge states, and "MCKT-w/o Forget" removes the forgetting process of knowledge states. The best results are highlighted in bold.

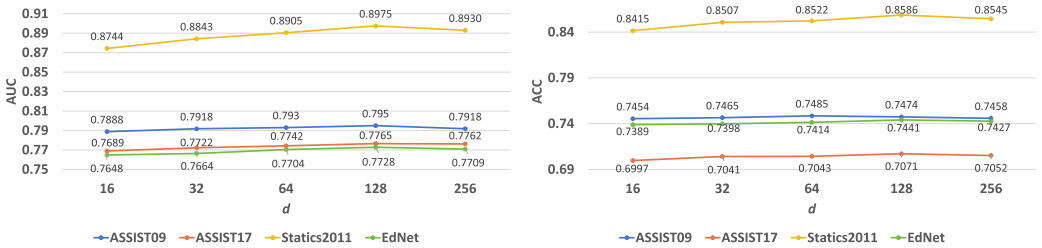


Fig. 5. The impact of feature embedding dimension size d on MCKT performance.

a student's knowledge state. (2) MCKT-w/o forget shows a similar degree of performance decline compared to MCKT. This suggests that considering the forgetting of a student's knowledge state during the answering process is effective. This aligns with the view that over time, students gradually forget what they have previously learned [2, 38, 46].

5.6 Sensitivity Analysis (RQ3)

In this section, we will explore the impact of hyperparameters on MCKT. Specifically, we investigate the feature embedding dimension size d, the similarity threshold δ, and the balance coefficients for total loss φ_1 and φ_2.

5.6.1 The Impact of d. We set $d \in [16, 32, 64, 128, 256]$ to investigate its impact on MCKT, and their performance is shown in Figure 5. We can draw the following conclusions: (1) Smaller values of d do not benefit the performance of MCKT, as observed in the case of $d = 16$. This phenomenon is attributed to the fact that representations of questions, interactions, and knowledge states all depend on the dimensionality size of embeddings d. A dimension that is too small cannot capture richer semantics to facilitate model optimization. (2) Larger values of d are advantageous for MCKT. It is evident that when $d = 128$, MCKT achieves optimal performance. When $d = 256$, MCKT also demonstrates good performance gains; however, it is important to note that a larger d implies greater computational resource consumption and further increases in model parameters, which may carry the risk of overfitting. In this article, we consistently set $d = 128$.

5.6.2 The Impact of δ. In this section, we will explore the impact of the similarity threshold δ (Equation (14)) on MCKT. Clearly, δ plays a crucial role in the construction of topic knowledge state. We select $\delta \in [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1]$ and examine their influence on MCKT performance, as shown in Figure 6. We draw the following conclusions: (1) Smaller values of δ do not benefit the performance of MCKT, as the performance is significantly worse for $\delta < 0.2$ compared to other cases. It is important to note that smaller δ implies that a greater number of

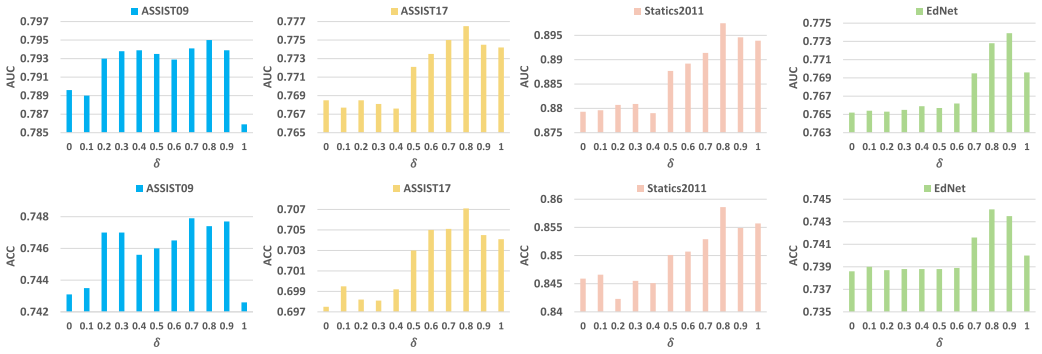


Fig. 6. The impact of different values of δ on MCKT performance across all datasets.

Table 5. The Impact of Removing the Similarity Threshold δ on MCKT, where "MCKT-w/o Sim" Indicates the Removal of the Constraint for Constructing Topic Knowledge State Similarity Thresholds

Model	ASSIST09		ASSIST17		Statics2011		EdNet	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
MCKT-w/o sim	0.7899	0.7432	0.7674	0.6999	0.8803	0.8472	0.7648	0.7385
MCKT	**0.7950**	**0.7474**	**0.7765**	**0.7071**	**0.8975**	**0.8586**	**0.7728**	**0.7441**

The best results are highlighted in bold.

interaction history are used in building topic knowledge state, but the performance deteriorates for smaller δ values. This emphasizes the importance of setting an appropriate similarity threshold when constructing topic knowledge state. (2) Overall, the performance of MCKT first increases as δ increases, and then decreases as δ increases. This phenomenon suggests the following: first, as mentioned in (1), smaller δ values are not beneficial for MCKT, as they weaken the role of topic knowledge state by using too many interaction history. Secondly, excessively large δ (e.g., $\delta = 1$) are also not advantageous for MCKT, as setting a very high similarity threshold results in using too few interaction history, leading to the construction of poor topic knowledge state. (3) When $\delta \in [0.7, 0.8, 0.9]$, the performance of MCKT is optimal. This is in line with expectations because a moderate δ uses an appropriate amount of interaction history. Notably, a performance stability is observed at $\delta = 0.8$, indicating that when constructing topic knowledge state, information from interaction history with a similarity of $\geq$0.8 to the current question is particularly crucial.

We conducted experiments to validate the effectiveness of setting the similarity threshold δ for the topic knowledge state. As shown in Table 5, MCKT-w/o sim removed the constraint of the similarity threshold, meaning that the topic knowledge state, in this case, is constructed from the entire interaction history, similar to the domain knowledge state. We observed that when the constraint of the similarity threshold was removed, the performance of MCKT-w/o sim significantly decreased compared to MCKT. Although in this case, the information used for constructing the topic knowledge state expanded to encompass the complete interaction history, it failed to adapt to the domain knowledge state, resulting in a decline in performance. This illustrates the effectiveness of the topic knowledge state constructed in this article, indicating that interaction history aligning with the "topic" holds significance for the topic knowledge state.

5.6.3 The Impact of φ_1 and φ_2. In this section, we will explore the impact of φ_1 and φ_2 on MCKT, where φ_1 influences the question-level and interaction-level contrastive loss, and φ_2 affects the

Table 6. Performance of the Total Loss Balance Coefficients φ_1 and φ_2 on the ASSIST09 Dataset

ASSIST09	$\varphi_2 = 0.0001$		$\varphi_2 = 0.001$		$\varphi_2 = 0.01$		$\varphi_2 = 0.1$		$\varphi_2 = 1$	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
$\varphi_1 = 0.0001$	0.7687	0.7285	0.7669	0.7290	0.7671	0.7287	0.7670	0.7270	0.7395	0.7158
$\varphi_1 = 0.001$	0.7684	0.7286	0.7689	0.7296	0.7677	0.7292	0.7668	0.7291	0.7423	0.7191
$\varphi_1 = 0.01$	0.7684	0.7246	0.7678	0.7286	0.7692	0.7285	0.7673	0.7284	0.7382	0.7152
$\varphi_1 = 0.1$	0.7696	0.7310	0.7701	0.7303	0.7691	0.7316	0.7687	0.7315	0.7402	0.7147
$\varphi_1 = 1$	0.7939	0.7474	0.7936	**0.7481**	**0.7950**	0.7474	0.7903	0.7443	0.7785	0.7375

We chose $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$. The best results are highlighted in bold.

Table 7. Performance of the Total Loss Balance Coefficients φ_1 and φ_2 on the ASSIST17 Dataset

ASSIST17	$\varphi_2 = 0.0001$		$\varphi_2 = 0.001$		$\varphi_2 = 0.01$		$\varphi_2 = 0.1$		$\varphi_2 = 1$	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
$\varphi_1 = 0.0001$	0.7700	0.7003	0.7706	0.7013	0.7757	0.7060	**0.7765**	**0.7071**	0.7691	0.7004
$\varphi_1 = 0.001$	0.7708	0.7014	0.7697	0.7016	0.7754	0.7053	0.7757	0.7072	0.7691	0.6995
$\varphi_1 = 0.01$	0.7706	0.7005	0.7703	0.7018	0.7744	0.7048	0.7752	0.7050	0.7713	0.7020
$\varphi_1 = 0.1$	0.7693	0.7010	0.7687	0.7005	0.7715	0.7034	0.7735	0.7037	0.7699	0.7002
$\varphi_1 = 1$	0.7653	0.6966	0.7663	0.6978	0.7678	0.6998	0.7690	0.7012	0.7643	0.6966

We chose $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$. The best results are highlighted in bold.

knowledge state-level contrastive loss. We set $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$, and investigate their performance on four datasets, as shown in Tables 6–9. In each case, we have highlighted the best AUC and ACC values in bold. We can draw the following conclusions: (1) Regarding the question-level and interaction-level contrastive loss, on the ASSIST09 dataset, larger φ_1 values, such as $\varphi_1 = 1$, tend to be more beneficial in promoting MCKT. This is likely because the ASSIST09 dataset is highly sensitive to the relationships between questions. On other datasets, a moderate φ_1 value, such as $\varphi_1 = 0.01$, benefits the model, and the performance shows a trend of first increasing and then decreasing as φ_1 increases. Too small φ_1 limit the advantages of question and interaction representations, leading to restricted model performance. Excessively large φ_1, except for the ASSIST09 dataset, are not conducive to model optimization because they make the model overly focus on question and interaction representations, making KT task optimization more challenging. (2) Regarding the knowledge state-level contrastive loss, as φ_2 increases, MCKT's performance on all datasets generally follows an increasing-then-decreasing trend. Initially, too small φ_2 fail to allow the model to capture the similarities and differences between domain knowledge states and topic knowledge states. Subsequently, too large φ_2 make the model overly focus on differentiating domain knowledge states from topic knowledge states, which is not conducive to model training. When $\varphi_2 = 0.1$, MCKT's performance achieves a relatively stable advantage.

5.7 $T + N$ Prediction (RQ4)

In this section, we conducted $T + N$ prediction experiments [34, 77] to explore the stability of MCKT in predicting future student performance. Specifically, let the current time be T. This experiment involved not only predicting a student's performance at the next timestep $(T + 1)$, but also predicting

Table 8. Performance of the Total Loss Balance Coefficients φ_1 and φ_2 on the Statics2011 Dataset

Statics2011	$\varphi_2 = 0.0001$		$\varphi_2 = 0.001$		$\varphi_2 = 0.01$		$\varphi_2 = 0.1$		$\varphi_2 = 1$	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
$\varphi_1 = 0.0001$	0.8923	0.8540	0.8933	0.8536	0.8950	0.8560	0.8949	0.8551	0.8899	0.8505
$\varphi_1 = 0.001$	0.8940	0.8549	0.8940	0.8549	0.8959	0.8549	0.8961	0.8557	0.8888	0.8508
$\varphi_1 = 0.01$	0.8940	0.8543	0.8940	0.8553	0.8956	0.8564	**0.8975**	**0.8586**	0.8903	0.8520
$\varphi_1 = 0.1$	0.8926	0.8550	0.8928	0.8529	0.8951	0.8554	0.8942	0.8561	0.8891	0.8520
$\varphi_1 = 1$	0.8853	0.8491	0.8875	0.8499	0.8880	0.8504	0.8889	0.8517	0.8858	0.8497

We chose $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$. The best results are highlighted in bold.

Table 9. Performance of the Total Loss Balance Coefficients φ_1 and φ_2 on the EdNet Dataset

EdNet	$\varphi_2 = 0.0001$		$\varphi_2 = 0.001$		$\varphi_2 = 0.01$		$\varphi_2 = 0.1$		$\varphi_2 = 1$	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
$\varphi_1 = 0.0001$	0.7717	0.7425	0.7714	0.7428	0.7722	0.7428	0.7720	0.7423	0.7703	0.7425
$\varphi_1 = 0.001$	0.7714	0.7414	0.7716	0.7424	0.7720	0.7408	0.7716	0.7418	0.7698	0.7418
$\varphi_1 = 0.01$	0.7717	0.7419	0.7725	0.7428	0.7723	0.7431	**0.7728**	**0.7441**	0.7692	0.7418
$\varphi_1 = 0.1$	0.7680	0.7407	0.7688	0.7407	0.7685	0.7406	0.7671	0.7397	0.7688	0.7415
$\varphi_1 = 1$	0.7658	0.7401	0.7656	0.7397	0.7667	0.7404	0.7651	0.7404	0.7587	0.7368

We chose $\varphi_1, \varphi_2 \in [0.0001, 0.001, 0.01, 0.1, 1]$. The best results are highlighted in bold.

the student's performance at $T + 2$, $T + 3$, ..., $T + N$ (still using the knowledge state at time $T + 1$). Clearly, as N increases, if the model can maintain good performance, it indicates the model's stable ability to predict future learning outcomes. We compared MCKT with some high-performing models (as shown in Table 2, including AKT, CDKT, DIMKT, and QIKT) in the $T + N$ prediction experiments, and their performance is depicted in Figures 7 and 8. We can find: (1) As N increases, all models exhibit a noticeable decrease in performance. This is expected because the KT task is primarily focused on predicting a student's performance at $T + 1$, and attempting to predict their performance at $T + 2$, $T + 3$, ..., $T + N$ inevitably weakens the model's ability. (2) The performance of attention-based KT models (such as AKT and MCKT in this article) has dropped significantly. In fact, the attention-based KT model (including MCKT) is not suitable for $T + N$ prediction experiments, because the knowledge state they construct often depends on the question at $T + 1$ (because it uses the question as the query to construct the knowledge state), and this knowledge state obviously does not apply to questions at $T + 2$, $T + 3$,..., $T + N$. (3) Despite MCKT not being specifically designed for this experiment (given that its topic knowledge state is constructed with attention), due to the advantages of having two types of knowledge states, MCKT still benefits from the domain knowledge state and knowledge state-level contrastive loss. We found that MCKT maintains its performance advantage in almost all scenarios, indicating its impressive stability.

5.8 The Time Complexity of MCKT (RQ5)

To gain a deeper understanding of MCKT, this section analyzes its time complexity. Specifically, we begin with a theoretical analysis of the time complexity of MCKT's core components, followed by experimental measurements of its actual execution time across different datasets.

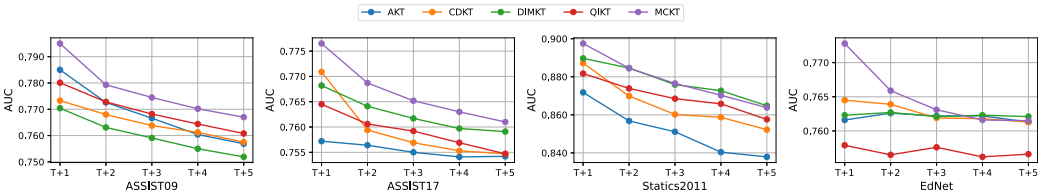


Fig. 7. $T + N$ prediction experiments on all datasets for MCKT and some high-performing KT models. We compare their performance (AUC) at timesteps $T + 1$ to $T + 5$.

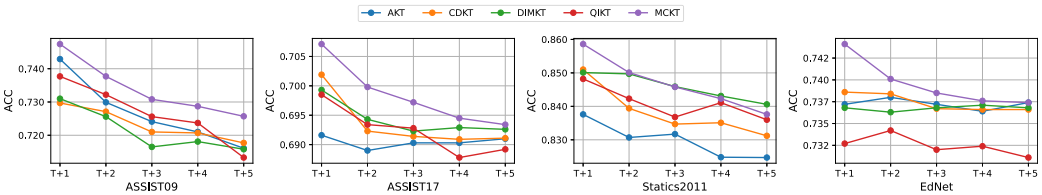


Fig. 8. $T + N$ prediction experiments on all datasets for MCKT and some high-performing KT models. We compare their performance (ACC) at timesteps $T + 1$ to $T + 5$.

5.8.1 Theoretical Analysis. As mentioned previously, the number of questions is denoted as n, the maximum number of timesteps in the sequence is F, and the embedding dimension is d. Since the knowledge state contrastive loss is computed within a mini-batch, the batch size is denoted as B. The time complexity of MCKT involves several key components:

- —Question-level Contrastive Loss: This involves calculating the similarity between question representations, with a complexity of $O(n^2d)$.
- —Interaction-level Contrastive Loss: This involves calculating the similarity between positive interaction representations (the representations of correctly answered questions), with a complexity of $O(n^2d)$.
- —Domain Knowledge State: This is computed based on an LSTM model, with a complexity of $O(Fd^2)$.
- —Topic Knowledge State: This is computed using a Transformer model, with a complexity of $O(F^2d + Fd^2)$.
- —Knowledge State-level Contrastive Loss: This involves calculations across F timesteps, with each timestep having a complexity of $O(B^2d)$, leading to a total complexity of $O(FB^2d)$.
- —Model Prediction: This involves calculations over F timesteps, where each step corresponds to an MLP operation with a complexity of $O(d^2)$, resulting in a total complexity of $O(Fd^2)$.

Since the embedding layer only involves lookup operations, its time complexity is omitted from the overall calculation.

As a result, the overall time complexity of MCKT is $O(n^2d) + O(Fd^2) + O(F^2d) + O(FB^2d)$.

5.8.2 Experimental Measurements. To explore the actual time consumption (in minutes) of MCKT, we report its data across all datasets. For comparison, in Table 10, we present the training time (Time/epoch) and model convergence time (Total time) [73] for MCKT and four commonly used classic baselines (the representative model of each group). We can observe the following: First, MCKT has a smaller training time compared to DKVMN and GKT, indicating that MCKT outperforms them both in terms of performance and efficiency. Second, DKT has the smallest training time, as it is solely composed of an LSTM, which also leads to its highly limited performance.

Table 10. Consuming Training Time (in Minutes) for Different Models on All Datasets

Model	Model Group	ASSIST09		ASSIST17		Statics2011		EdNet	
		Time/epoch	Total time	Time/epoch	Total time	Time/epoch	Total time	Time/epoch	Total time
DKT [43]	Sequential based	**0.05**	**1.85**	**0.02**	**0.56**	**0.01**	**0.33**	**0.13**	**3.90**
DKVMN [79]	Memory based	0.08	10.40	0.03	2.64	0.02	2.77	0.20	8.20
GKT [39]	Graph based	1.28	52.48	1.11	92.58	0.29	24.07	17.92	501.76
AKT [16]	Attention based	0.08	2.24	0.04	2.80	0.01	1.12	0.22	21.78
MCKT	Sequential based+Attention based	0.22	4.62	0.04	2.52	0.02	1.76	0.45	5.88

The best results are bold.

Table 11. The Execution Time (in Minutes) of Key Stages in MCKT (Contrastive Learning and Inference)

Stage	ASSIST09 Time/epoch	ASSIST17 Time/epoch	Statics2011 Time/epoch	EdNet Time/epoch
MCKT-contrastive learning	0.10	0.02	0.01	0.18
MCKT-inference	0.03	0.01	0.01	0.06

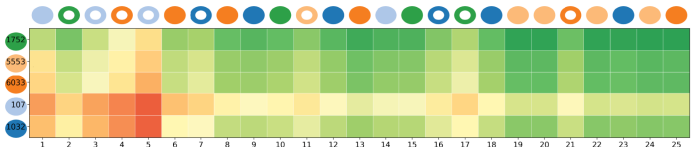
Additionally, although MCKT requires more time per epoch than AKT, MCKT converges more quickly. Finally, in terms of absolute time, MCKT's training time is relatively short while achieving the best performance, further demonstrating the effectiveness of MCKT.

We also report the time consumption of two key stages in a training epoch of MCKT: Contrastive learning and inference. As shown in Table 11, MCKT-contrastive learning refers to the time of computing three contrastive losses in one training epoch, while MCKT-inference refers to the time consumption of testing the model each time. According to Table 11, the time consumption of MCKT in both key stages is very small. This indicates that MCKT maintains low computational costs while achieving efficient contrastive learning and inference, further enhancing its practical feasibility and efficiency for real-world use.

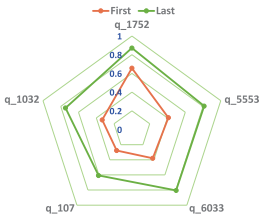
6 Application

One of the most interesting applications of KT is probably monitoring students' knowledge states. Once a student's knowledge state is accurately captured, teachers can provide targeted guidance to address their weaknesses in understanding. In this section, we randomly selected a student's answer records for five questions over 25 timesteps from the ASSIST09 dataset and visualized the student's knowledge state for each question at each timestep.

As shown in Figure 9(a), the y-axis represents the questions that the student answered, with different questions distinguished by different colored circles. The x-axis represents the student's interactions (questions and responses), with correct answers shown as the corresponding circles and incorrect answers having an additional white circle. Each element in the matrix represents the student's mastery of the question at that moment, with larger values indicating a better understanding. We can draw the following conclusions: (1) When a student answers a question correctly/incorrectly, their mastery of that question significantly increases/decreases, which aligns with intuition because correctness in answering questions does reflect a student's knowledge level for that specific question. (2) We can notice that as time progresses, students gradually forget their previous knowledge. For example, the student's mastery of question 6033 is high during timesteps 8–13 but significantly decreases at timesteps 16–17. This emphasizes the importance of considering students' forgetting during the learning process. (3) Moreover, in the initial stages of tracing, a



(a) An example of the evolution of a student's knowledge state over 25 time steps for 5 questions in the ASSIST09 dataset. The y-axis represents specific questions, which are distinguished by different colored circles. The x-axis represents student interactions (questions and responses), showing the corresponding circle if the student answered correctly. If the student answered a question incorrectly, an additional white circle is added to indicate the difference. Each element in the matrix represents the student's mastery of that particular question at the corresponding time step.



(b) The radar chart displays the student's mastery levels on 5 questions at the first time step and the last time step.

Fig. 9. An application of KT.

student's knowledge state is not stable, but with more practice over time, the traced knowledge state becomes increasingly stable. (4) Finally, as shown in Figure 9(b), the student's mastery of all questions significantly improves from the first timestep to the last timestep, even if the student made mistakes during the learning process. This is consistent with real-life scenarios, where errors can also enhance knowledge, and continuous practice of questions benefits students in improving their knowledge mastery.

7 Conclusions and Future Work

In this article, we propose a novel Multi-level Contrastive learning for Knowledge Tracing (MCKT). Unlike relying on data augmentation strategies, MCKT delves deep into domain knowledge and leverages contrastive learning at three distinct levels to acquire representations conducive to KT: question-level, interaction-level, and knowledge state-level. Furthermore, we model student learning data from different perspectives to comprehensively capture their knowledge state. Extensive experiments on four publicly available datasets demonstrate that MCKT outperforms other KT models significantly. Ablation experiments confirm the effectiveness of multi-level contrastive learning.

Of course, MCKT still offers room for further improvements. The way question-level and interaction-level contrastive learning selects positive samples relies on the concept labels behind the questions. While using concept labels in questions is a prevalent practice in current KT models, exploring methods to overcome this limitation is a promising direction. Additionally, MCKT lacks interpretability, although this is a common issue in most KT models, a more decision-transparent KT model would certainly be more desirable. Lastly, the complex relationships between questions, interactions, and knowledge states can be further explored using graph structures to more effectively capture the semantic connections between them. In future work, we will investigate how to address MCKT's reliance on concept labels using unsupervised methods. Simultaneously, we will introduce additional interpretable components to enhance the interpretability of MCKT, such as attention scores and IRT predictions. Furthermore, we plan to apply GCL to delve deeper into the complex relationships between questions, interactions, and knowledge states by modeling them as graph nodes. By leveraging methods like graph attention networks, we aim to learn the dependencies between them and further uncover the relationships between questions and interactions, questions and knowledge states, as well as interactions and knowledge states. This will improve MCKT's ability to model contextual information and semantic connections, thereby enhancing its overall performance.

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